

A4: Machine Learning - Exam Lab



This exam practice page is designed for the A4: Machine Learning unit of the new IB Computer Science syllabus (first assessment 2027).

Syllabus Links

[A4.1 Machine learning fundamentals eBook](#)

[A4.2 Data preprocessing \(HL Only\) eBook](#)

[A4.3 Machine learning approaches \(HL only\) eBook](#)

[A4.4 Ethical considerations eBook](#)

The Exam Scenario

Scenario: GreenFlow is a technology firm that develops machine learning solutions for commercial farms to optimise crop yields and minimise water waste. They use a variety of sensors and high-resolution cameras to monitor plant health. To deploy these models effectively across various farm environments, ranging from small family plots to massive industrial operations, GreenFlow must carefully consider hardware configurations and the ethics of its automated decision-making processes.

Core Questions (HL and SL)

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 1: Machine Learning Fundamentals (SL & HL)

(a) **Describe** two different approaches to machine learning that GreenFlow might use, providing a specific application for each within the farm scenario. **[4 marks]**

(b) GreenFlow needs to deploy their models on both mobile edge devices (like drones) and central cloud servers. **Describe** the hardware requirements for these two different deployment scenarios. **[4 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

Question 2: Ethical Considerations (SL & HL)

(a) **Discuss** the ethical implications of GreenFlow's machine learning models in terms of **algorithmic fairness** and **accountability**. **[6 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

HL Only: Advanced Machine Learning

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 3: Data Preprocessing (HL Only)

(a) GreenFlow's sensors often produce "noisy" data due to environmental interference. **Describe** the significance of **data cleaning** in this context and **outline** one technique for handling **outliers**. **[4 marks]**

(b) **Explain** the importance of **feature selection** in reducing the **curse of dimensionality** for GreenFlow's plant health model. **[3 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

Question 4: Advanced Approaches & Evaluation (HL Only)

(a) GreenFlow is comparing **Decision Trees** and **K-Nearest Neighbours (K-NN)** for classifying soil quality. **Explain** how a Decision Tree would categorise a new soil sample compared to the K-NN approach. **[4 marks]**

(b) GreenFlow prioritises identifying *every* diseased plant, even if it occasionally misidentifies a healthy one as sick. **Explain** why they might focus on **recall** rather than **precision** as an evaluation metric in this specific situation. **[3 marks]**

Level 1: Strategy

Level 2: Mark Scheme

Level 3: Worked Example

How confident do you feel with this topic?

Exam Ready: I answered most questions correctly without looking at the eBook. I can explain these concepts to someone else and understand the "why" behind the mark scheme.

Move on to the next topic.

Practitioner: I understood the general concepts, but I missed a few marks on specific terminology or complex applications. I needed the "Strategy" guide for the harder questions.

Annotate your answers with the Mark Scheme and revisit the eBook sections where you lost marks. You can also use the bookmarking and notes feature on the eBook pages to highlight certain areas.

Novice: I struggled to get started on the questions. I found it difficult to apply the theory to the practice scenarios.

Go back to the eBook and Worked Examples before attempting these questions again. Perhaps use some of the Quizlets to practice the key terms and review the command terms.

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